SC4020 Tasks 2 and 3 — executed evidence

This experiment holds out canonical ligand–sequence pairs while keeping every evaluation entity in training. Seed 2026; per-ligand 70/15/15 allocation with pre-freeze target-coverage repair. Final counts and repair moves are in data/processed/splits/task2/*/manifest.json. The official brief and accompanying design plan were unavailable; the supplied agenda governs this implementation.

Data and scales

dataset	raw_ligands	raw_targets	raw_finite	unique_ligands	unique_targets	unique_pairs	missing_fraction
davis	68	442	30056	68	379	25772	0.0000
kiba	2111	229	118254	2068	229	117657	0.7516

DAVIS converts positive raw Kd in nM once using pKd=9−log10(Kd). Keep screening-limit measurements; any-censored flags and censored fractions survive mean aggregation. A merged score can include censored and uncensored repeats, so a score different from 5 can still have a censored source. KIBA uses supplied transformed scores unchanged. Canonical isomeric SMILES and exact normalized sequences define identity; pair repeats are mean-merged after conversion, with source cells/IDs, sample std and screening fractions saved.

Methods and selection

Global, ligand and target means use warm-train only. Additive bias regularization is selected from {1,10,50} by validation. Target kNN uses overlap-restricted cosine on training-only additive-bias residuals, shrinkage n_common/(n_common+beta), K={5,10,20,50}, beta={0,10,50,100}, and positive available neighbours; no usable neighbour falls back to additive bias. MF is mu+b_ligand+b_target+U·V, trained on observed pairs by our NumPy/Numba SGD (no recommender package). The per-observation SGD objective includes L2 factor/bias penalties, hence a degree-weighted total regularizer. Ranks {0,4,8,16,32}, reg {.01,.1}, learning rate .02, at most 160 epochs and patience 16 use validation MSE, with validation concordance as trial tie-breaker. Rank zero is an SGD bias-only ablation. Predictions are not clipped; training bounds are saved. No test score selects parameters or epochs.

Test comparison

dataset	model	mse	ci_global	ci_per_ligand	hit1	hit5	hit10	mrr
davis	global_mean	0.6432	0.5000	0.5000	0.0178	0.0889	0.1776	0.0819
davis	ligand_mean	0.4843	0.7679	0.5000	0.0178	0.0889	0.1776	0.0819
davis	target_mean	0.5993	0.6385	0.6985	0.0746	0.3881	0.5522	0.2368
davis	bias_only	0.4321	0.8052	0.6981	0.1045	0.3881	0.5522	0.2554
davis	target_knn	0.2465	0.8654	0.8134	0.3284	0.6716	0.8209	0.4827
davis	MF	0.2559	0.8654	0.8021	0.4030	0.6418	0.8060	0.5276
davis	random_ranking	—	—	0.5000	0.0178	0.0889	0.1776	0.0819
kiba	global_mean	0.7123	0.5000	0.5000	0.2222	0.4603	0.6828	0.4234
kiba	ligand_mean	0.4250	0.7075	0.5000	0.2222	0.4603	0.6828	0.4234
kiba	target_mean	0.6177	0.6807	0.6767	0.3781	0.6805	0.8637	0.5645

dataset	model	mse	ci_global	ci_per_ligand	hit1	hit5	hit10	mrr
kiba	bias_only	0.3531	0.7854	0.6848	0.3910	0.6879	0.8718	0.5737
kiba	target_knn	0.2190	0.8419	0.7633	0.5300	0.8476	0.9457	0.6923
kiba	MF	0.2012	0.8410	0.7687	0.5683	0.8726	0.9573	0.7209
kiba	random_ranking	—	—	0.5000	0.2222	0.4603	0.6828	0.4234

MSE uses observed labels only. Higher pKd and supplied transformed KIBA mean stronger affinity. CI denotes concordance: exclude true ties, award predicted ties half credit, compare with brute force in tests. Primary rankings recommend targets for each ligand within held-out measured candidates, with all true-best ties relevant. Hit and reciprocal rank are expected values over uniform ordering of predicted ties. Singleton and all-truth-tied groups are excluded and counted. Hit@5 requires >=5 candidates and Hit@10 requires >=10; hit5_gt5 further excludes the trivial five-candidate case. Per-target rankings/CI are a secondary interpretation of “recommend a drug for a target”. Random rankings use exact expectations, with no regression MSE. Candidate sets differ across datasets/splits, so raw Hit values are not directly comparable. No unobserved KIBA pairs are labelled negatives. Group bootstrap intervals describe uncertainty within a fixed split and are distinct from concordance and five-split standard deviation.

DAVIS: answers to the Task 2 questions

Does it learn beyond lookup? All evaluated test pairs were absent from training. MF test MSE is 0.255887, versus ligand mean 0.484315 and additive bias 0.432096; its relative reduction over bias is 40.78%. Pearson association of held-out true bias residuals with the latent dot product is 0.6370. This is descriptive evidence of interaction structure; MF also learns its own biases, so the dot-product correlation alone is not a causal ablation.

Is CF necessary? MF improves regression over the selected additive bias on this split. The training matrix density is 0.6992; the selected rank is 32, with 14752 parameters versus 448 for bias-only. Matrix completion is useful for deliberately withheld interactions, whereas observed complete DAVIS pairs can already be looked up. The parameter count/density does not establish clinical utility or superiority for unseen proteins.

Sparsity and abnormal values. Training co-observation quantiles (target pairs) are {'0.0': 16.0, '0.25': 30.0, '0.5': 33.0, '0.75': 36.0, '1.0': 48.0}; selected kNN is {'k': 5, 'beta': 100.0}, test fallback fraction 0.0000%. See the overlap distribution, sparsity support table, score-bin errors and screening audit below. Dense/sparse performance uses training observation counts only.

min_train_count	max_train_count	pairs	mf_mse	bias_mse	knn_fallback_fraction
100	100000	3876	0.2559	0.4321	0.0000

Case studies from reload. Three eligible ligands were chosen using suggested IDs when available, otherwise observed sparse/median/dense training histories. Only held-out measured targets are candidates. Top-10 needs at least ten candidates; no training labels are substituted.

original_drug_ids	train_observations	test_candidates	true_best_predicted_rank	true_best_bias_rank	mse	ci_global	hit1	hit5	mrr
44259	265	57	6	21	0.8889	0.7987	0.0000	0.0000	0.1667
11338033	265	57	4	46	0.1605	0.9697	0.0000	1.0000	0.2500
10127622	265	57	19	14	0.0497	0.6786	0.0000	0.0000	0.0526

Figures: `figures/task2_davis_comparison.png`, `figures/task2_davis_diagnostics.png`, `figures/task2_davis_learning_curve.png`. Tables: `tables/task2_davis_case_rankings.csv`, `tables/task2_davis_label_bins.csv`, `tables/task2_davis_diagnostics.csv`, `tables/task2_davis_overlap.csv`. Original-ID alias lists in case tables are the observed sources for that pair; entity tables contain the complete aliases.

KIBA: answers to the Task 2 questions

Does it learn beyond lookup? All evaluated test pairs were absent from training. MF test MSE is 0.201213, versus ligand mean 0.424951 and additive bias 0.353102; its relative reduction over bias is 43.02%. Pearson association of held-out true bias residuals with the latent dot product is 0.6371. This is descriptive evidence of interaction structure; MF also learns its own biases, so the dot-product correlation alone is not a causal ablation.

Is CF necessary? MF improves regression over the selected additive bias on this split. The training matrix density is 0.1719; the selected rank is 32, with 75802 parameters versus 2298 for bias-only. Matrix completion is useful for deliberately withheld interactions, whereas observed complete DAVIS pairs can already be looked up. The parameter count/density does not establish clinical utility or superiority for unseen proteins.

Sparsity and abnormal values. Training co-observation quantiles (target pairs) are {'0.0': 0.0, '0.25': 2.0, '0.5': 20.0, '0.75': 226.0, '1.0': 504.0}; selected kNN is {'k': 10, 'beta': 50.0}, test fallback fraction 0.1082%. See the overlap distribution, sparsity support table, score-bin errors and screening audit below. Dense/sparse performance uses training observation counts only.

min_train_count	max_train_count	pairs	mf_mse	bias_mse	knn_fallback_fraction
0	20	2657	0.4088	0.6281	0.0075
20	50	3161	0.2145	0.3456	0.0000
50	100	11280	0.1461	0.2854	0.0000
100	100000	1390	0.2213	0.3935	0.0000

Case studies from reload. Three eligible ligands were chosen using suggested IDs when available, otherwise observed sparse/median/dense training histories. Only held-out measured targets are candidates. Top-10 needs at least ten candidates; no training labels are substituted.

original_drug_ids	train_observations	test_candidates	true_best_predicted_rank	true_best_bias_rank	mse	ci_global	hit1	hit5	mrr
CHEMBL2004544	69	15	2	3	0.0991	0.7723	0.0000	1.0000	0.5000
CHEMBL565612	116	25	1	22	0.4856	0.6724	1.0000	1.0000	1.0000
CHEMBL2007574	42	10	1	9	0.0410	0.8140	1.0000	1.0000	1.0000

Figures: `figures/task2_kiba_comparison.png`, `figures/task2_kiba_diagnostics.png`, `figures/task2_kiba_learning_curve.png`. Tables: `tables/task2_kiba_case_rankings.csv`, `tables/task2_kiba_label_bins.csv`, `tables/task2_kiba_diagnostics.csv`, `tables/task2_kiba_overlap.csv`. Original-ID alias lists in case tables are the observed sources for that pair; entity tables contain the complete aliases.

Anomaly audit and scientific limits

The following finite nonpositive KIBA observations occur in the pinned upstream file. Their presence and provenance are verified; assay-level correctness cannot be established from these files. They remain flagged and retained, without relabelling missing values or deleting an inconvenient score.

drug_id	target_id	affinity	original_drug_id	original_target_id	source_cell	censored	suspicious
1323	118	0	CHEMBL373751	P53350	1336:118	False	True

DAVIS screening-bound values are benchmark labels, not exact measured affinities. Mean-merging aliases changes the usual benchmark counts and loses assay/isoform distinctions. This is a new canonical-identity split, not the official DeepDTA folds. Only one warm seed was used. Rankings within measured candidates do not establish whole-proteome retrieval.

Optional train-only thinning

fraction	model	training_pairs	unseen_test_pairs	mse	ci_global
1.0000	bias_only	18020	0	0.4321	0.8052
1.0000	MF	18020	0	0.2559	0.8654
0.5000	bias_only	9010	0	0.4384	0.8027
0.5000	MF	9010	0	0.3422	0.8340
0.2500	bias_only	4505	0	0.4504	0.7935
0.2500	MF	4505	0	0.4379	0.7938
0.1000	bias_only	1802	31	0.4914	0.7624
0.1000	MF	1802	31	0.4836	0.7612
0.0500	bias_only	901	363	0.5252	0.7360
0.0500	MF	901	363	0.5103	0.7357

Nested 100/50/25/10/5% subsets change training only; the full warm test is fixed. Full-training validation-selected hyperparameters are retained, with checkpoint epoch selected on the same validation partition for each subset. Unseen entity factors/biases are zero, giving partial/global additive fallback. Figure: `figures/task2_davis_thinning.png`.

Paired uncertainty

dataset	model	baseline	ligands	replicates	mse_difference	bootstrap_95_low	bootstrap_95_high	seed
davis	MF	bias_only	68	1000	-0.1762	-0.2316	-0.1248	2026
kiba	MF	bias_only	2068	1000	-0.1519	-0.1661	-0.1391	2026

Intervals use 1,000 paired ligand-group bootstrap replicates, and describe MF-minus-bias MSE on the fixed split. They do not account for split choice.

References and assistance

- Öztürk et al. (2018), DeepDTA, https://doi.org/10.1093/bioinformatics/bty593. Raw files from hkmztrk/DeepDTA commit a546a8433a6822e958f36171c4356ad6f414d623, with ordered mappings and SHA-256 in data/raw/*/source.json.
- DeepDTA data documentation: https://github.com/hkmztrk/DeepDTA/blob/master/data/README.md (raw Kd units, transformed KIBA, NaN missingness).
- He et al. (2017), SimBoost, https://doi.org/10.1186/s13321-017-0209-z (transformed KIBA direction and DAVIS screening-bound provenance).
- Biased MF is the standard additive-bias plus dot-product model. SGD, residual target-kNN, cold factor ridge, tie-expectation metrics and experiment orchestration are implemented locally. NumPy/SciPy/scikit-learn/RDKit/Numba provide numerical, fixed-feature and acceleration support; no pretrained recommender is used.
- Local reuse: CF SGD, Fenwick CI, 3-mer features and tests were adapted from ../drug_target_recommender/src and tests. That sibling implementation and Task 1 artifacts were preserved. Codex provided substantial implementation, analysis, testing and writing assistance. Human contributor names, contributions and the course's assistance policy must be checked by the authors; no identities or approval claims are invented.

Task 3 — cold-target report evidence (v1)

Five separately trained target-sequence holdouts use seeds 2026–2030. Approximately 70/15/15% of unique exact normalized sequences go to train/validation/test. Every affinity measurement for a sequence remains in one partition. All model selection uses cold validation only; test sequences have zero training affinity observations. Evaluation ligands without training observations are explicitly flagged and retain a global/partial-bias fallback; exact counts are in split manifests and saved prediction files.

Algorithms

C0: ligand means from each cold-training partition, with global training mean for unknown ligands. **C1:** compare a cold full sequence with training sequences using L2-normalized fixed A–Z 3-mer count cosine. Take top K training proteins; transfer available observations for that ligand, weighted by cosine^power, otherwise use C0. This is algebraically C0 plus weighted ligand-mean residuals. K={5,15,50}, power={1,4} are selected on cold validation. Missing ligand–neighbour measurements contribute no weight.

C2 learned factor ridge: fit a separate rank-16 biased MF for up to 80 fixed-budget epochs on cold-train only (train-error checkpoint, reg .01 and LR .02); never import warm-task factors. Represent sequences with fixed 3-mer, 20-AA composition and 400-dipeptide frequencies. Train-target-centered multioutput ridge maps these sequence features to learned training-target bias and factors. Tune ridge alpha {.1,1,10,100} on cold validation. Predict mu+b_ligand+predicted_bias+U_ligand·predicted_factor. Cold queries do not use target-ID factors. Unobserved ligand factors are zero. The MF optimizer seed is fixed at 2026 for all split seeds. Cached radius-2 2048-bit Morgan features are available; C2 uses cold-train ligand factors instead. No SVD, imputation, pretrained embedding or test-label fitting is used.

Neither dataset supplied an S-W similarity file locally, so this implementation computes label-free sequence cosine instead of downloading or assuming a normalization. Cosine is bounded 0..1, but is not alignment identity. Pre-specified <.3/.3–.5/.5–.7/>=.7 buckets describe this feature similarity; do not interpret them as sequence-identity family thresholds.

Five-split aggregate

dataset	model	mse_mean	mse_std	ci_global_mean	ci_global_std	hit1_mean	hit5_mean	mrr_mean
davis	factor_ridge	0.3374	0.0429	0.8365	0.0084	0.2657	0.5184	0.3953
davis	ligand_mean	0.4800	0.0711	0.7675	0.0076	0.0177	0.0885	0.0816
davis	sequence_neighbors	0.3025	0.0395	0.8448	0.0122	0.3407	0.6143	0.4645
kiba	factor_ridge	0.3240	0.0161	0.7564	0.0222	0.3614	0.6672	0.5380
kiba	ligand_mean	0.3868	0.0358	0.7077	0.0221	0.1969	0.4784	0.3918
kiba	sequence_neighbors	0.3061	0.0281	0.7694	0.0244	0.3755	0.7265	0.5573

The table gives arithmetic mean and sample standard deviation (ddof=1) across five repeated target splits; they are not independent five-fold CV, confidence intervals, or bootstrap variability. Every model is retrained per split. Full metric means/std/counts, including per-ligand and per-target concordance, are in tables/task3_aggregate.csv. Per-split primary model selection is saved before test evaluation and its aggregate is in tables/task3_validation_selected_aggregate.csv.

MSE uses observed labels only. Higher pKd and supplied transformed KIBA mean stronger affinity. CI denotes concordance: exclude true ties, award predicted ties half credit, compare with brute force in tests. Primary rankings recommend targets for each ligand within held-out measured candidates, with all true-best ties relevant. Hit and reciprocal rank are expected values over uniform ordering of predicted ties. Singleton and all-truth-tied groups are excluded and counted. Hit@5 requires >=5 candidates and Hit@10 requires >=10; hit5_gt5 further excludes the trivial five-candidate case. Per-target rankings/CI are a secondary interpretation of “recommend a drug for a target”. Random rankings use exact expectations, with no regression MSE. Candidate sets differ across datasets/splits, so raw Hit values are not directly comparable. No unobserved KIBA pairs are labelled negatives. Group bootstrap intervals describe uncertainty within a fixed split and are distinct from concordance and five-split standard deviation.

Per-split results

dataset	seed	model	selected_by_validation	mse	ci_global	ci_per_ligand	ci_per_target	hit1	hit5	mrr
davis	2026	ligand_mean	False	0.3969	0.7682	0.5000	0.8009	0.0175	0.0877	0.0812
davis	2026	sequence_neighbors	True	0.2794	0.8327	0.7405	0.8638	0.3030	0.5152	0.4018
davis	2026	factor_ridge	False	0.2860	0.8288	0.7121	0.8596	0.2273	0.4697	0.3535
davis	2027	ligand_mean	False	0.4441	0.7658	0.5000	0.8053	0.0178	0.0889	0.0819
davis	2027	sequence_neighbors	True	0.2793	0.8498	0.7612	0.8718	0.2794	0.6471	0.4396
davis	2027	factor_ridge	False	0.3285	0.8264	0.7229	0.8585	0.1618	0.4706	0.3012
davis	2028	ligand_mean	False	0.4995	0.7639	0.5000	0.8011	0.0175	0.0877	0.0812
davis	2028	sequence_neighbors	True	0.3161	0.8325	0.7527	0.8505	0.3333	0.5606	0.4441
davis	2028	factor_ridge	False	0.3374	0.8406	0.7688	0.8511	0.3030	0.4848	0.4160
davis	2029	ligand_mean	False	0.5876	0.7595	0.5000	0.8069	0.0175	0.0877	0.0812
davis	2029	sequence_neighbors	True	0.3661	0.8479	0.7604	0.8729	0.2879	0.5909	0.4166
davis	2029	factor_ridge	False	0.4050	0.8407	0.7642	0.8618	0.2424	0.5303	0.3797
davis	2030	ligand_mean	False	0.4721	0.7799	0.5000	0.8097	0.0181	0.0902	0.0827
davis	2030	sequence_neighbors	True	0.2718	0.8611	0.7871	0.8750	0.5000	0.7576	0.6206
davis	2030	factor_ridge	False	0.3301	0.8457	0.7464	0.8630	0.3939	0.6364	0.5264
kiba	2026	ligand_mean	False	0.4395	0.7010	0.5000	0.7656	0.2163	0.5013	0.4216
kiba	2026	sequence_neighbors	False	0.3220	0.7845	0.6656	0.8106	0.4192	0.7973	0.6135
kiba	2026	factor_ridge	True	0.3309	0.7802	0.6726	0.8036	0.4567	0.8017	0.6318
kiba	2027	ligand_mean	False	0.3826	0.7226	0.5000	0.8069	0.2249	0.5020	0.4240

dataset	seed	model	selected_by_validation	mse	ci_global	ci_per_ligand	ci_per_target	hit1	hit5	mrr
kiba	2027	sequence_neighbors	True	0.2783	0.7993	0.6898	0.8168	0.4385	0.7827	0.6185
kiba	2027	factor_ridge	False	0.3169	0.7678	0.6824	0.8244	0.4117	0.7042	0.5934
kiba	2028	ligand_mean	False	0.3952	0.6780	0.5000	0.7543	0.2534	0.5124	0.4583
kiba	2028	sequence_neighbors	True	0.3213	0.7373	0.6314	0.7820	0.4178	0.7556	0.6056
kiba	2028	factor_ridge	False	0.3017	0.7408	0.6663	0.7933	0.4330	0.7067	0.6107
kiba	2029	ligand_mean	False	0.3760	0.7354	0.5000	0.7494	0.1512	0.4613	0.3383
kiba	2029	sequence_neighbors	True	0.3351	0.7717	0.6180	0.7896	0.2638	0.5959	0.4409
kiba	2029	factor_ridge	False	0.3451	0.7672	0.6023	0.7764	0.2520	0.5547	0.4285
kiba	2030	ligand_mean	False	0.3405	0.7014	0.5000	0.7710	0.1384	0.4148	0.3169
kiba	2030	sequence_neighbors	True	0.2735	0.7543	0.6386	0.8014	0.3380	0.7009	0.5079
kiba	2030	factor_ridge	False	0.3252	0.7261	0.5820	0.7912	0.2537	0.5686	0.4255

Coverage, costs and failure analysis

dataset	seed	primary_model	knn_test_fallback_fraction	unseen_ligand_test_pairs	cold_mf_seconds	whole_run_seconds
davis	2026	sequence_neighbors	0.0000	0	0.2360	5.4425
davis	2027	sequence_neighbors	0.0000	0	0.2262	4.1375
davis	2028	sequence_neighbors	0.0000	0	0.2277	4.1803
davis	2029	sequence_neighbors	0.0000	0	0.2338	4.1206
davis	2030	sequence_neighbors	0.0000	0	0.2320	4.0985
kiba	2026	factor_ridge	0.0015	0	0.4964	6.5873
kiba	2027	sequence_neighbors	0.0036	0	0.5042	5.8722
kiba	2028	sequence_neighbors	0.0009	0	0.4970	5.7696
kiba	2029	sequence_neighbors	0.0019	0	0.5477	5.7960
kiba	2030	sequence_neighbors	0.0009	0	0.4790	5.8702

Sequence neighbours may have no observed measurement for a query ligand, especially in sparse KIBA; C0 then provides no target-specific adjustment. The factor-ridge method supplies an adjustment through sequence features, but a linear sequence-to-latent mapping can miss target selectivity and overfit small training target sets. Composition and 3-mers lose order beyond three residues; variable full sequence lengths and domains alter cosine. Distribution shift and related sequences remaining across splits limit claims of remote-homology generalization. Per-target errors, sequence lengths, neighbour support and nearest-training similarities are exported for each split; bucket counts must accompany any performance claim. Empty buckets remain empty.

Files: tables/task3_{dataset}_seed{seed}_similarity_buckets.csv, *_per_target.csv and *_sequence_audit.csv; figures/task3{dataset}_seed{seed}_similarity.png and *_comparison.png. Saved neighbours and ridge archives contain training-only target sequences, observations or latent outputs, provenance, parameters and split hashes. Every archive was checked in a fresh interpreter with label-free queries, and identical score order after reload.

Scope and limitations

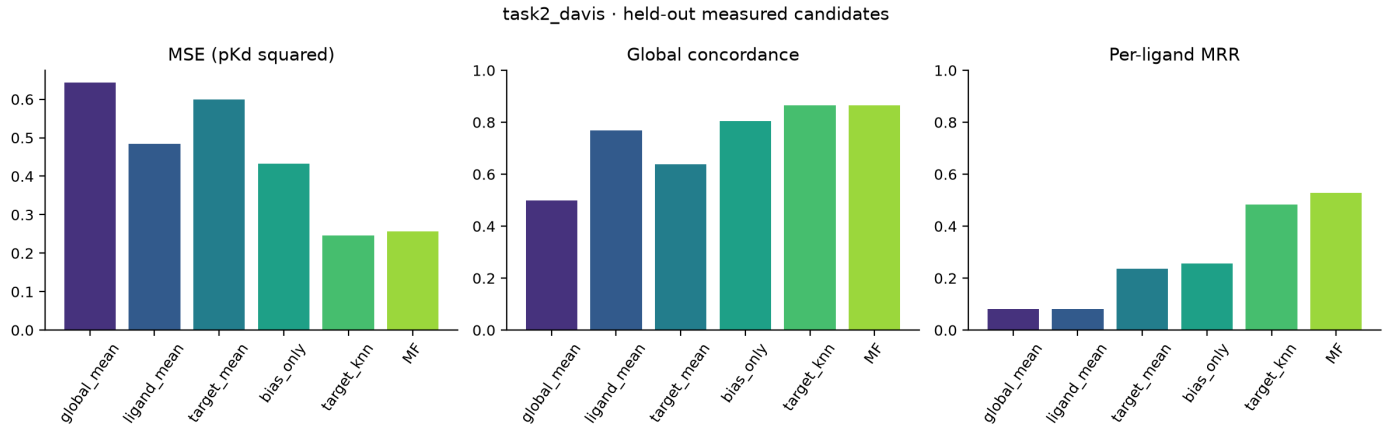
All core and priority-one models/splits/diagnostics in the agenda are implemented. The optional C3 pair regressor, ESM-2, strict family cold split, mixup/ensemble and censored-loss MF were not run; no results are asserted for them. Original SW matrices were not supplied and were not assumed available. The official rubric and project plan were not found, so exact rubric compliance beyond the agenda cannot be certified. Warm-pair and cold-target errors answer different questions and are not compared as the same protocol.

References and assistance

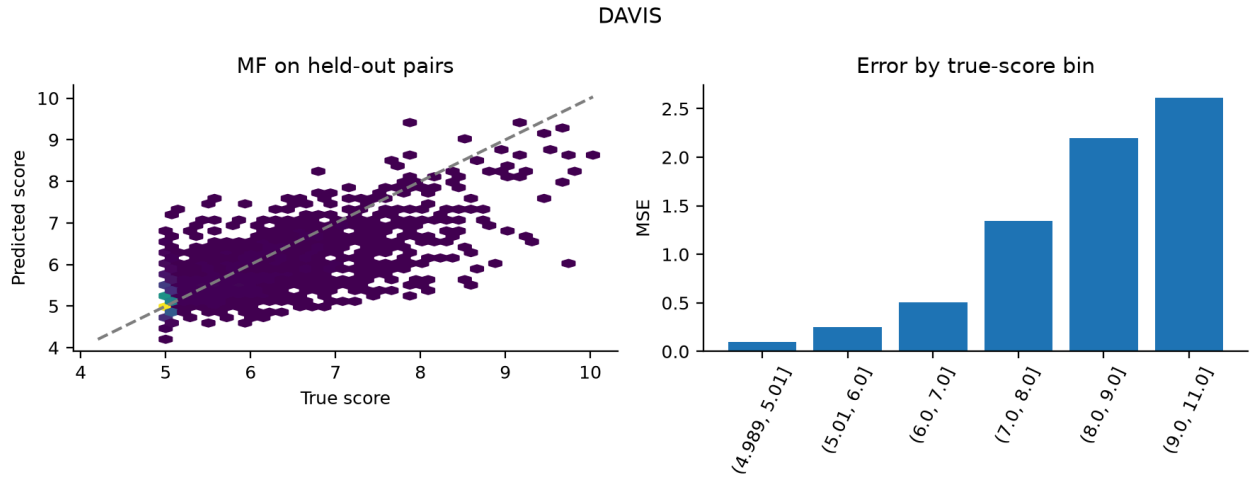
- Öztürk et al. (2018), DeepDTA, https://doi.org/10.1093/bioinformatics/bty593. Raw files from hkmztrk/DeepDTA commit a546a8433a6822e958f36171c4356ad6f414d623, with ordered mappings and SHA-256 in data/raw/*/source.json.
- DeepDTA data documentation: https://github.com/hkmztrk/DeepDTA/blob/master/data/README.md (raw Kd units, transformed KIBA, NaN missingness).
- He et al. (2017), SimBoost, https://doi.org/10.1186/s13321-017-0209-z (transformed KIBA direction and DAVIS screening-bound provenance).

- Biased MF is the standard additive-bias plus dot-product model. SGD, residual target-kNN, cold factor ridge, tie-expectation metrics and experiment orchestration are implemented locally. NumPy/SciPy/scikit-learn/RDKit/Numba provide numerical, fixed-feature and acceleration support; no pretrained recommender is used.
- Local reuse: CF SGD, Fenwick CI, 3-mer features and tests were adapted from ../drug_target_recommender/src and tests. That sibling implementation and Task 1 artifacts were preserved. Codex provided substantial implementation, analysis, testing and writing assistance. Human contributor names, contributions and the course's assistance policy must be checked by the authors; no identities or approval claims are invented.

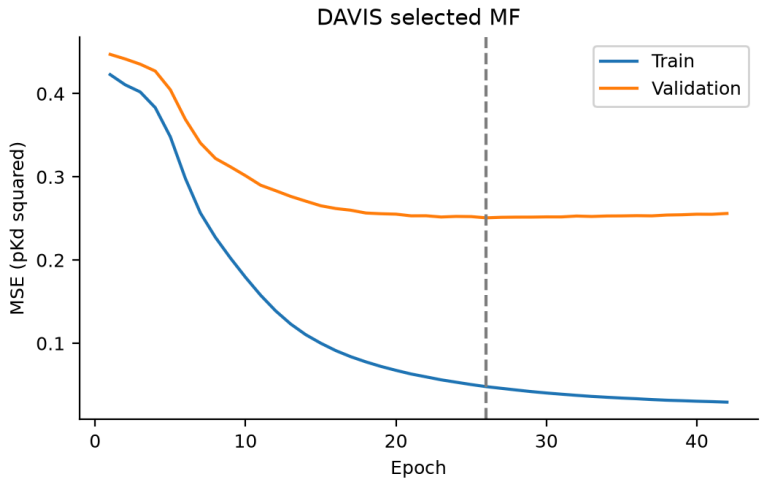
Task 2 figures — DAVIS



DAVIS — warm-pair model comparison

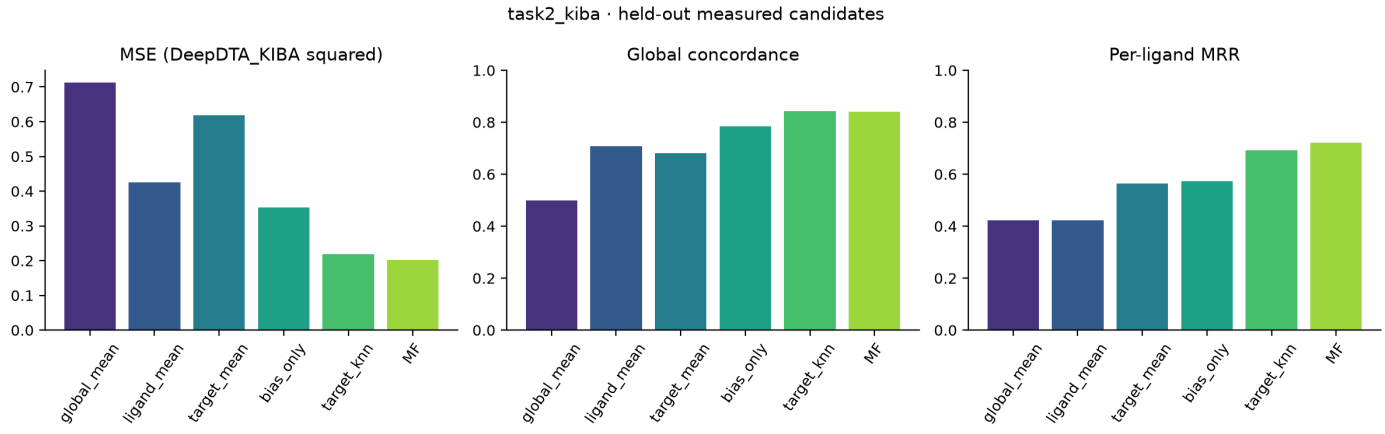


DAVIS — prediction and support diagnostics

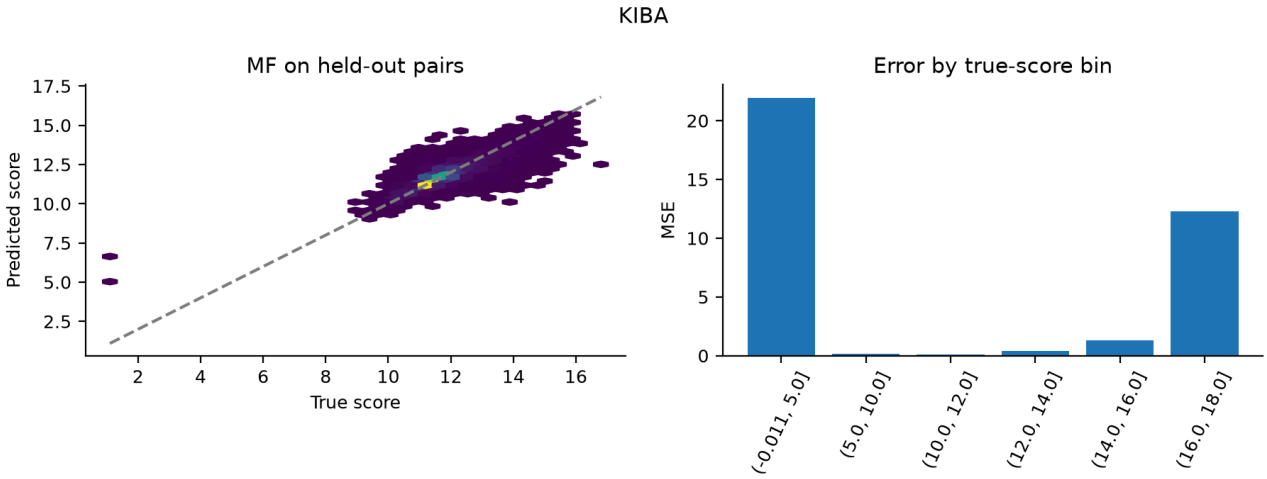


DAVIS — MF learning curve

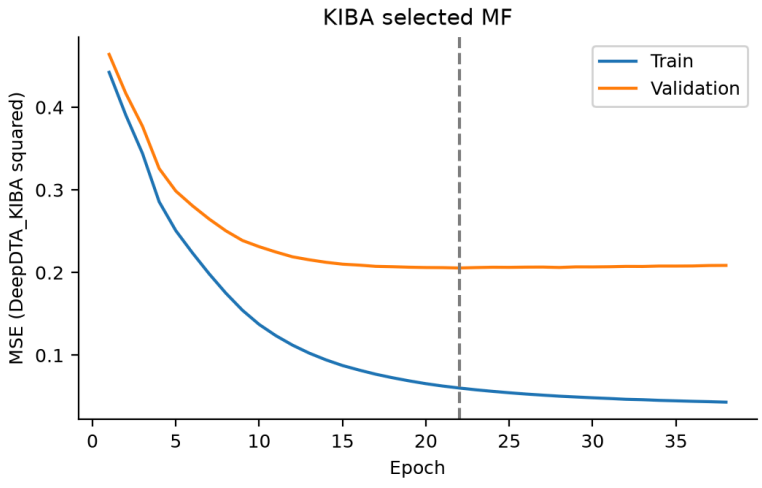
Task 2 figures — KIBA



KIBA — warm-pair model comparison

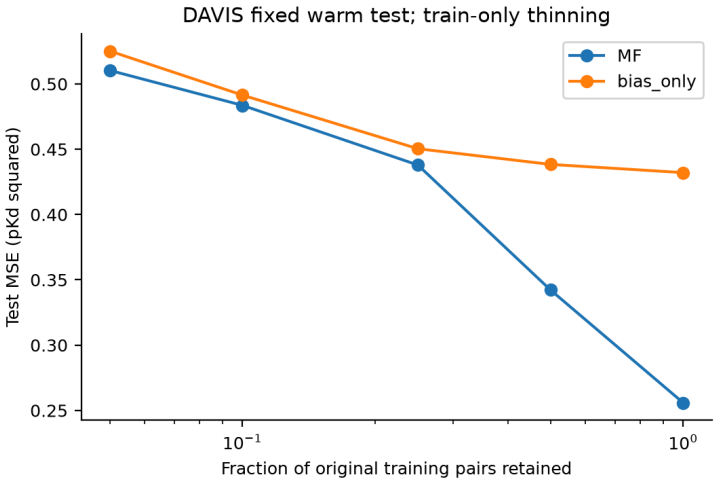


KIBA — prediction and support diagnostics



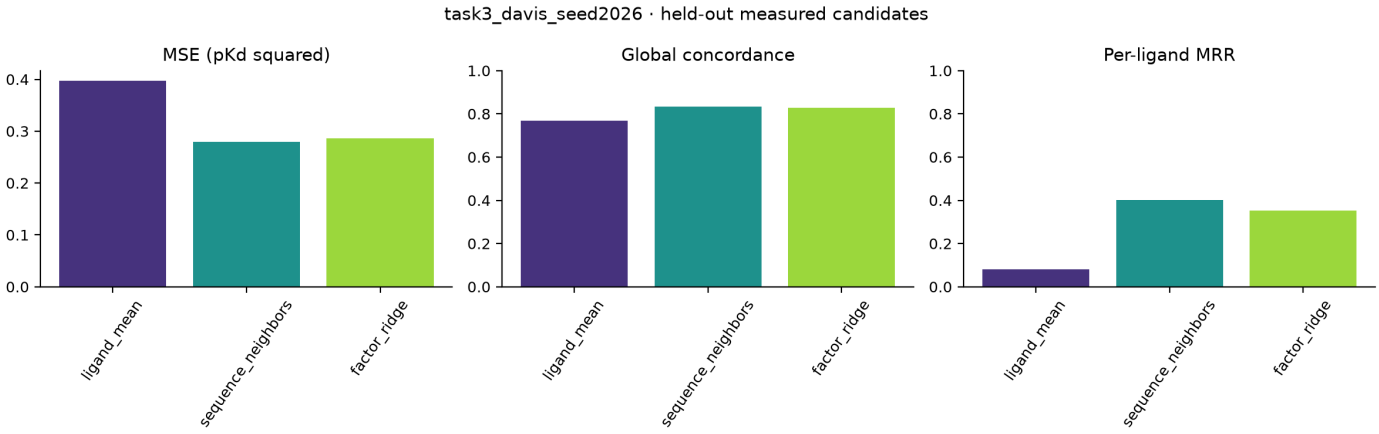
KIBA — MF learning curve

Task 2 — training-data thinning

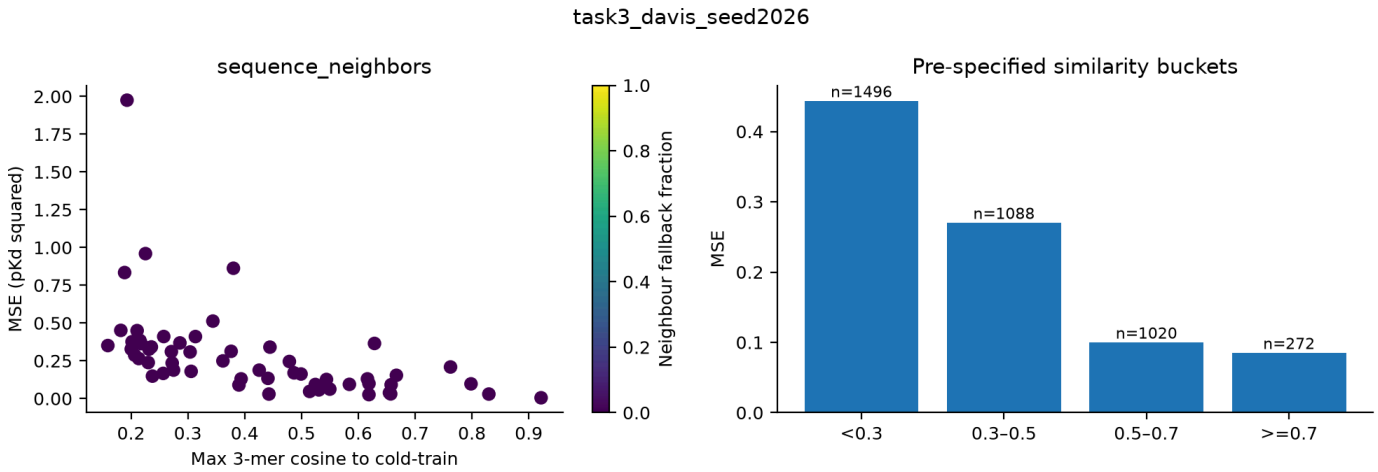


DAVIS — fixed-test performance as training observations decrease

Task 3 figures — DAVIS, seed 2026

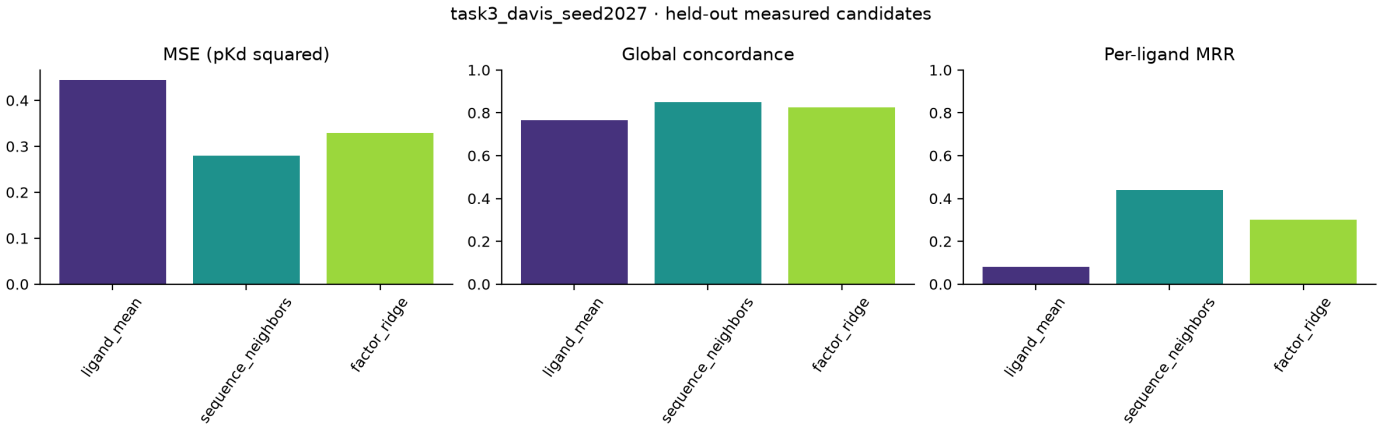


DAVIS, seed 2026 — cold-target model comparison

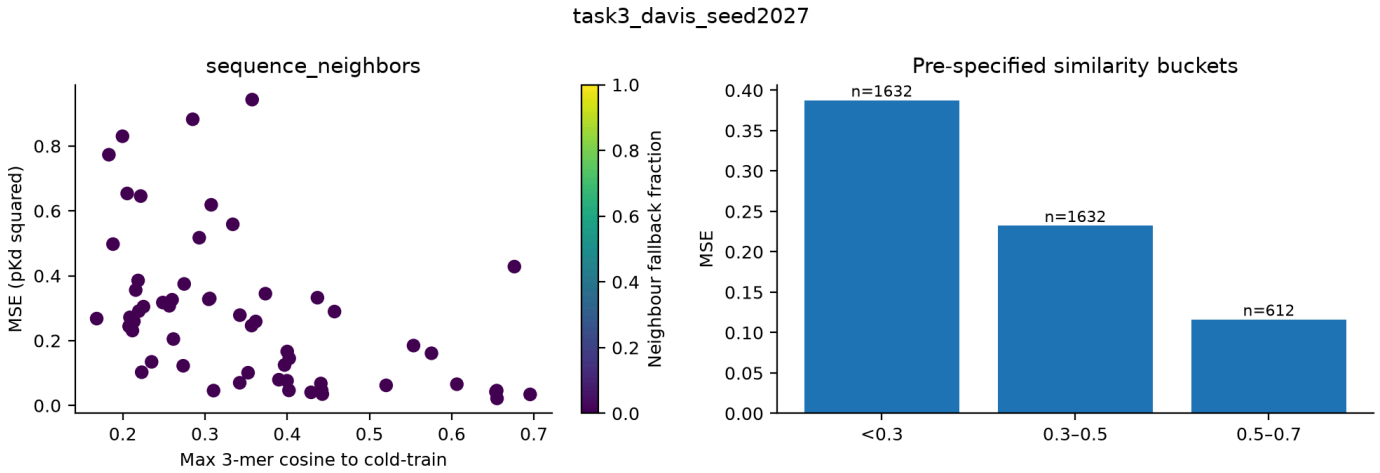


DAVIS, seed 2026 — similarity and error diagnostics

Task 3 figures — DAVIS, seed 2027

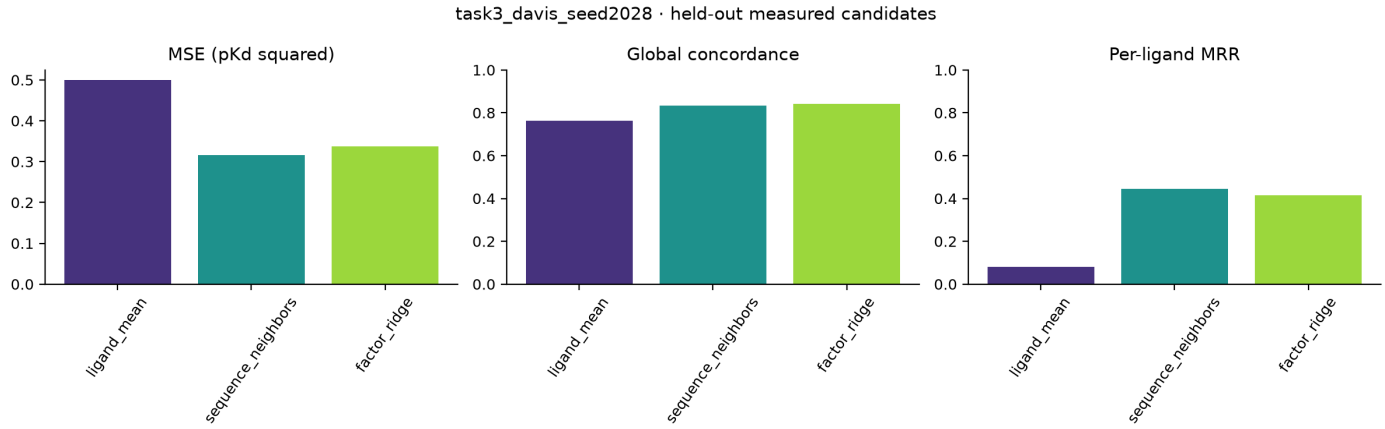


DAVIS, seed 2027 — cold-target model comparison

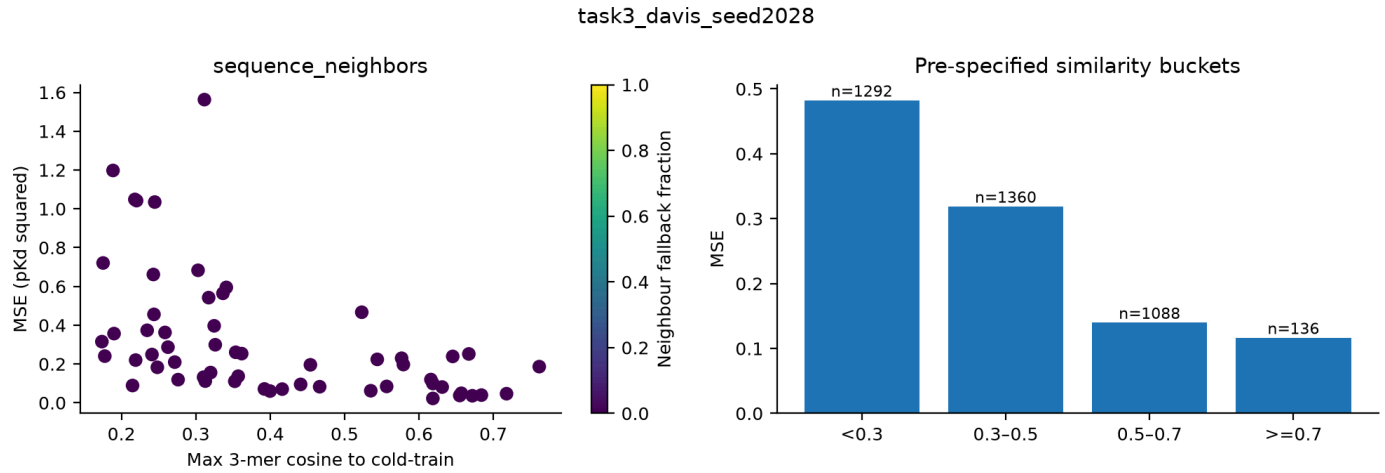


DAVIS, seed 2027 — similarity and error diagnostics

Task 3 figures — DAVIS, seed 2028

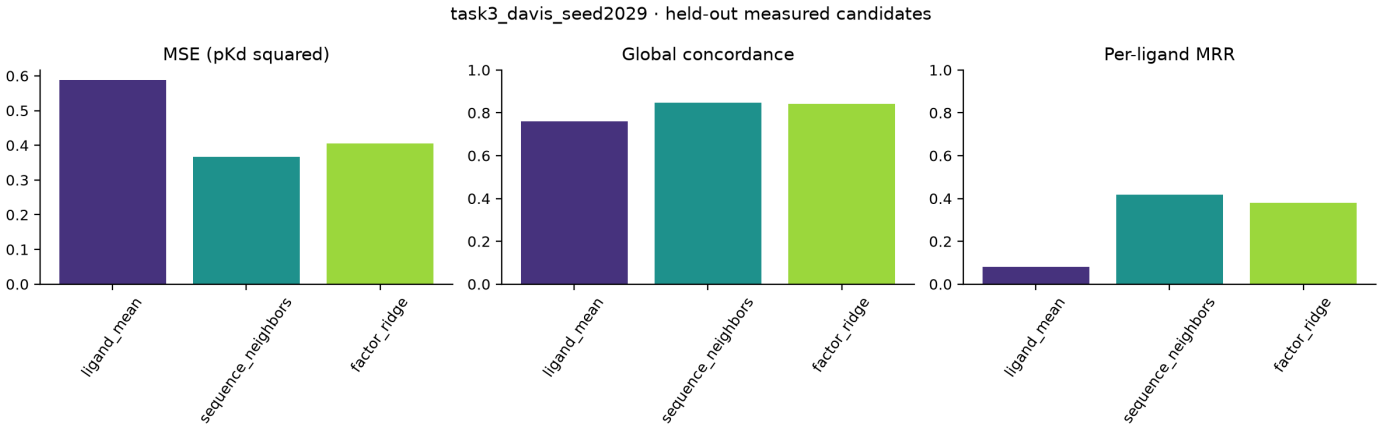


DAVIS, seed 2028 — cold-target model comparison

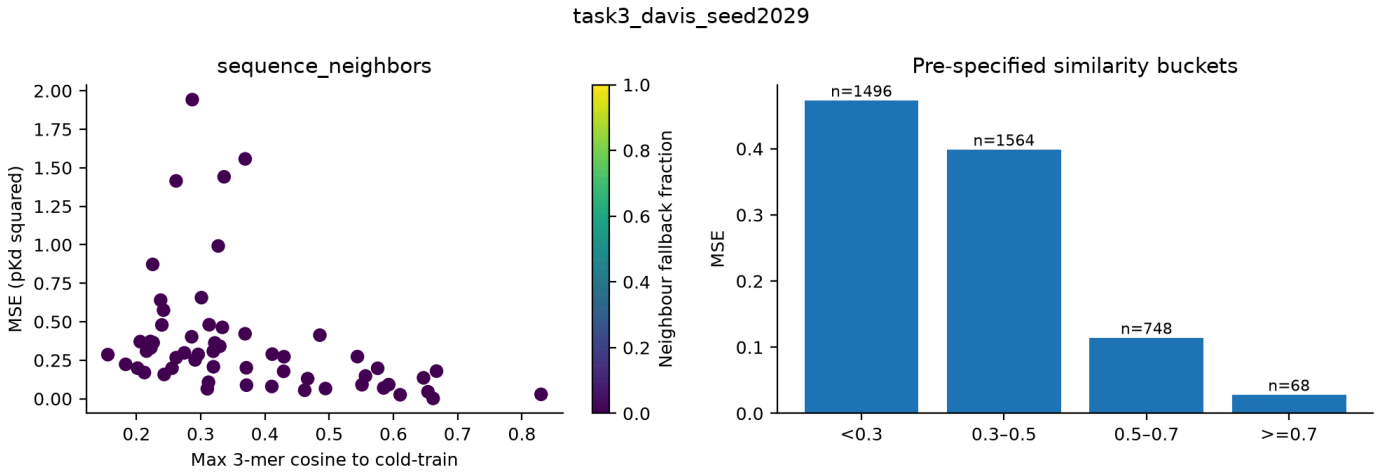


DAVIS, seed 2028 — similarity and error diagnostics

Task 3 figures — DAVIS, seed 2029

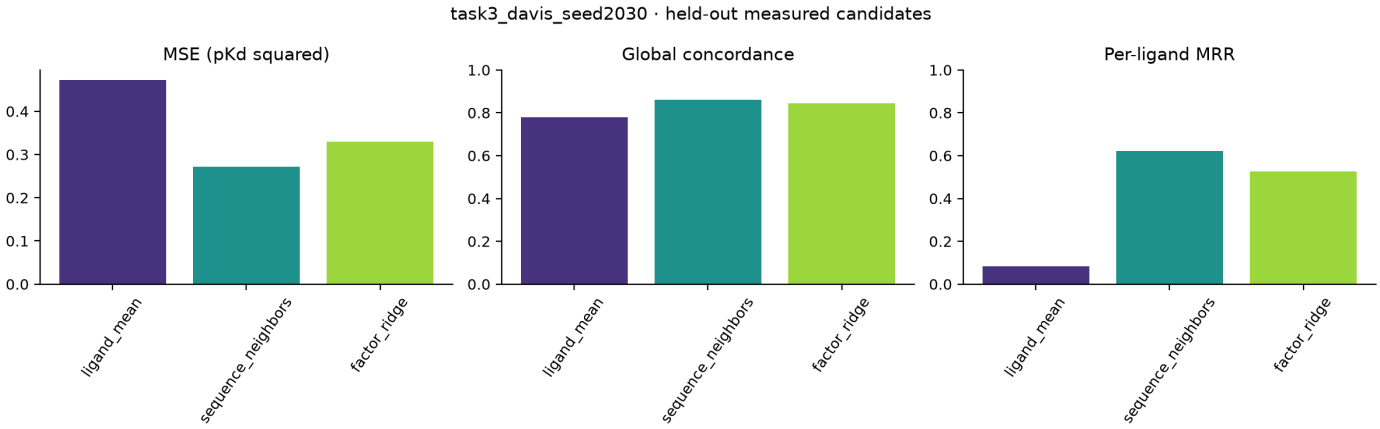


DAVIS, seed 2029 — cold-target model comparison

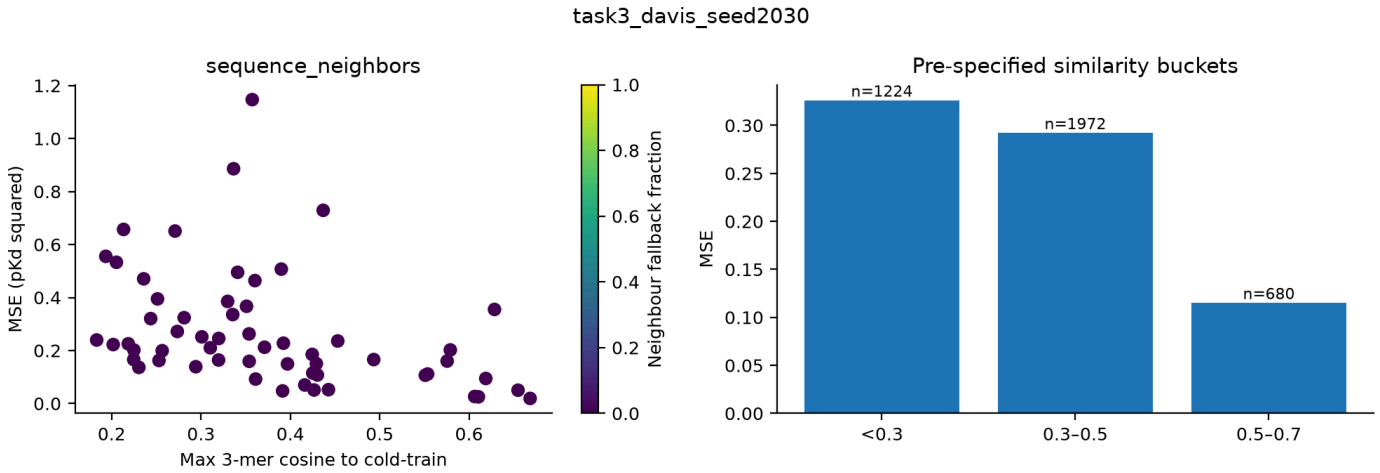


DAVIS, seed 2029 — similarity and error diagnostics

Task 3 figures — DAVIS, seed 2030

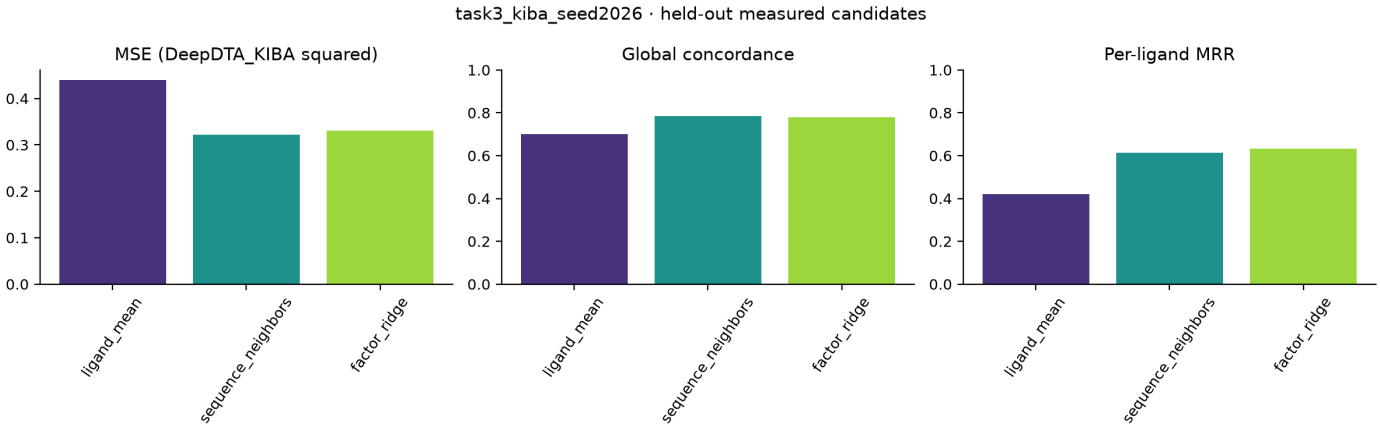


DAVIS, seed 2030 — cold-target model comparison

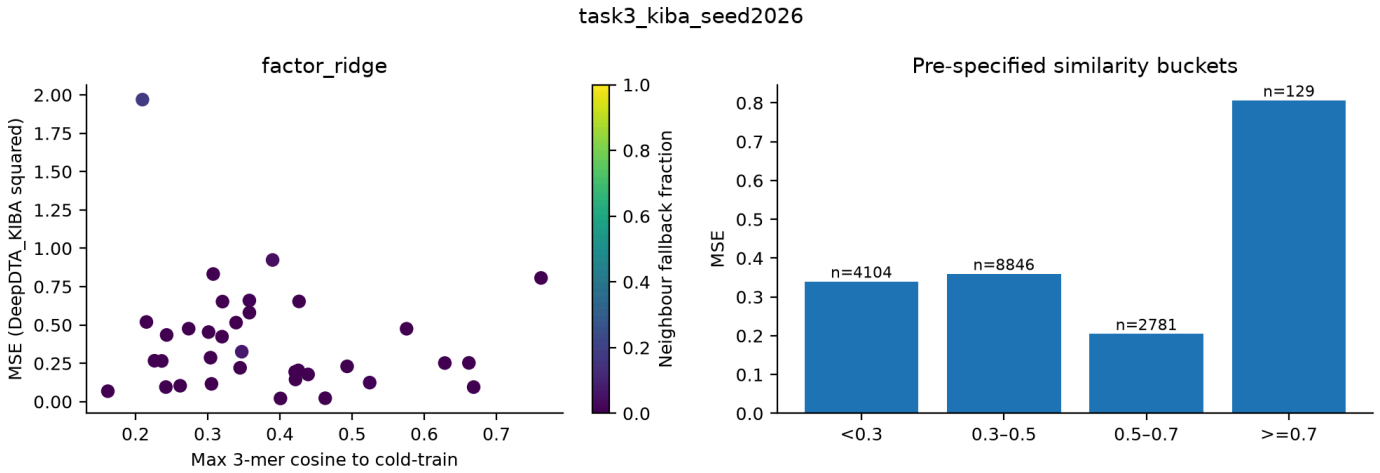


DAVIS, seed 2030 — similarity and error diagnostics

Task 3 figures — KIBA, seed 2026

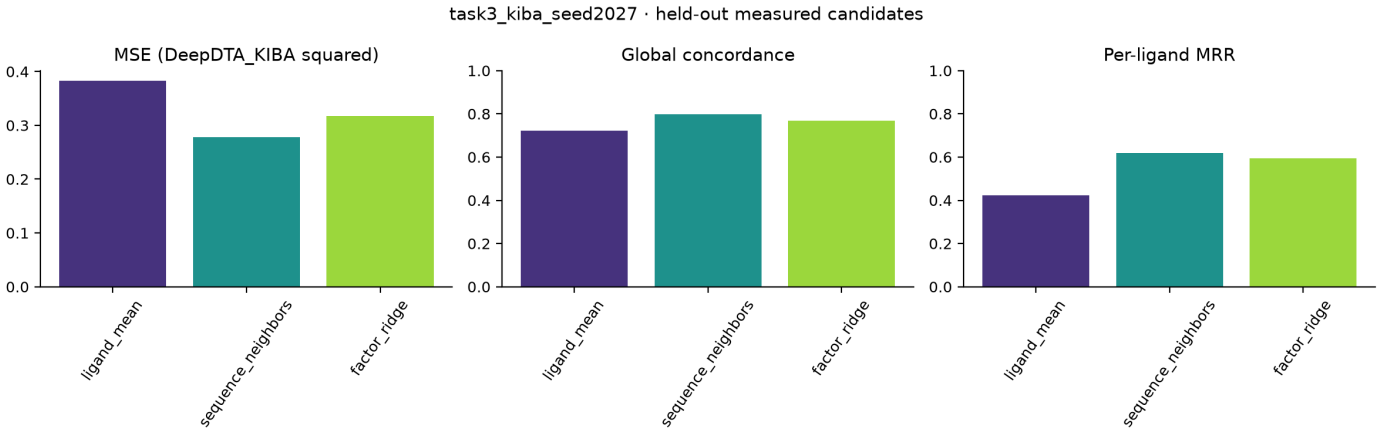


KIBA, seed 2026 — cold-target model comparison

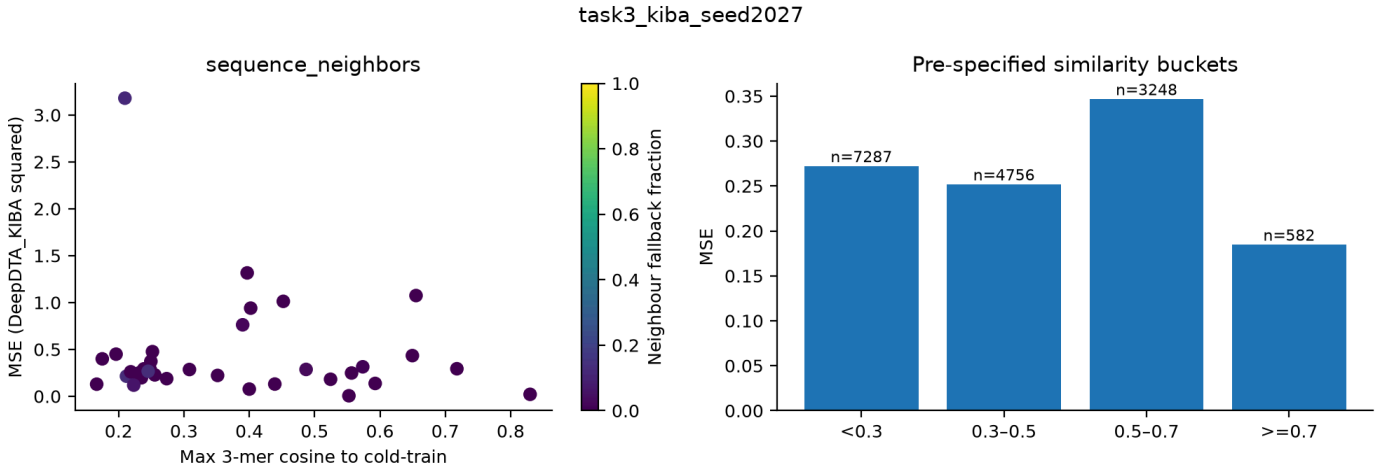


KIBA, seed 2026 — similarity and error diagnostics

Task 3 figures — KIBA, seed 2027

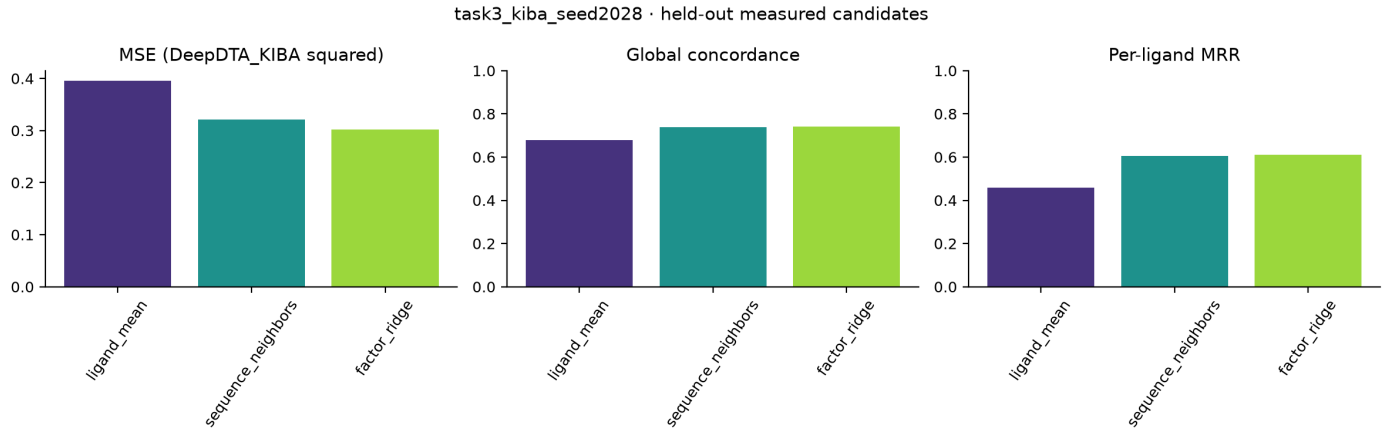


KIBA, seed 2027 — cold-target model comparison

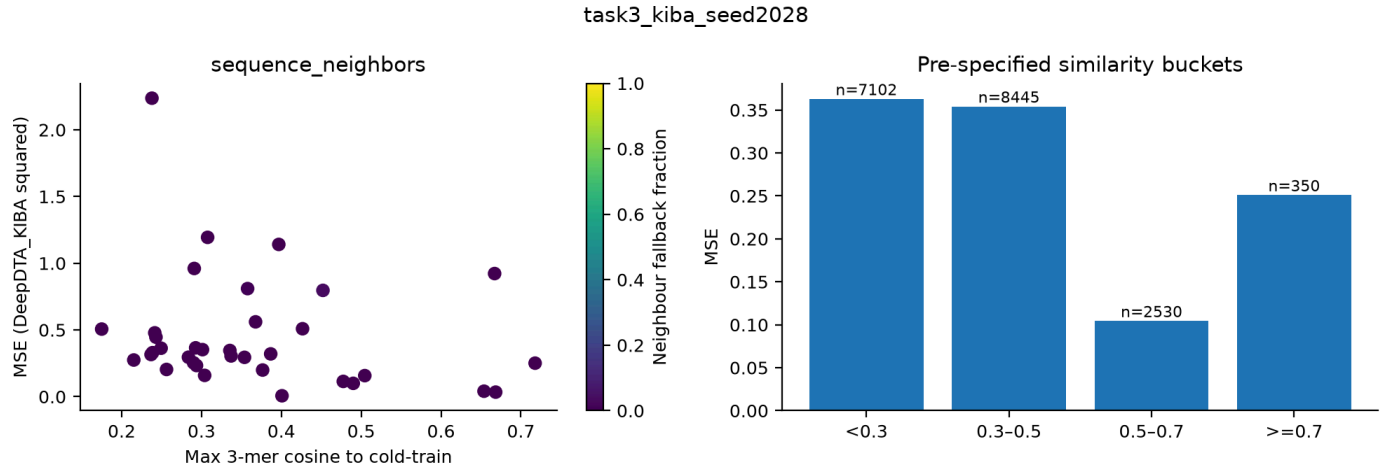


KIBA, seed 2027 — similarity and error diagnostics

Task 3 figures — KIBA, seed 2028

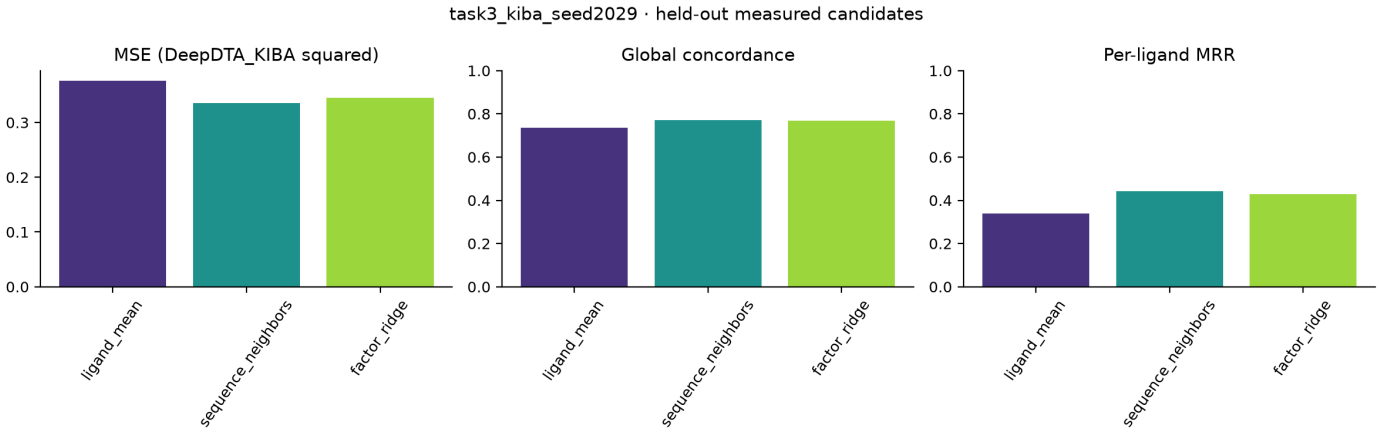


KIBA, seed 2028 — cold-target model comparison

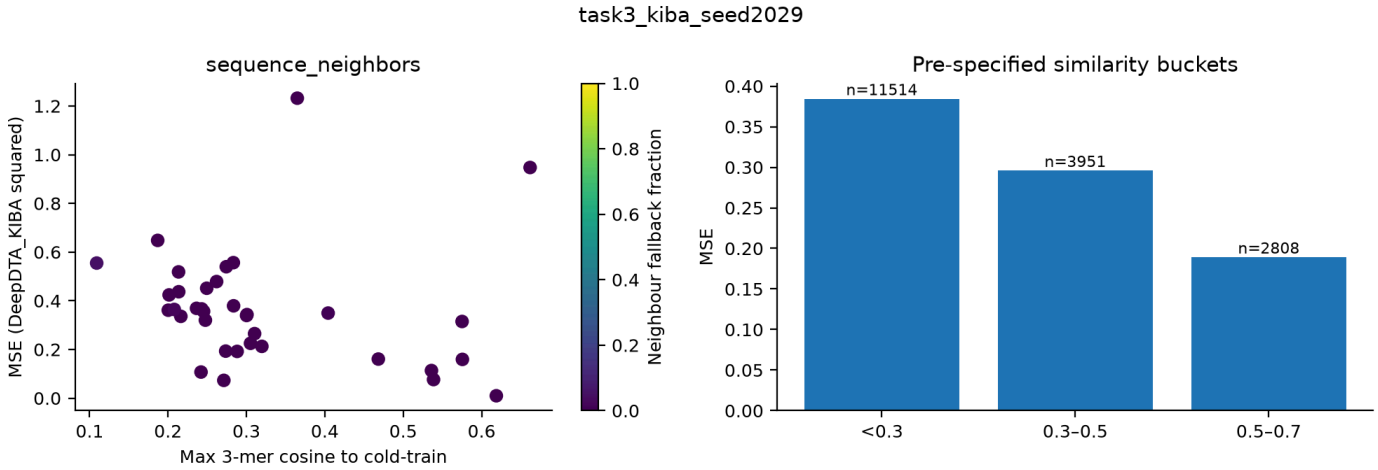


KIBA, seed 2028 — similarity and error diagnostics

Task 3 figures — KIBA, seed 2029

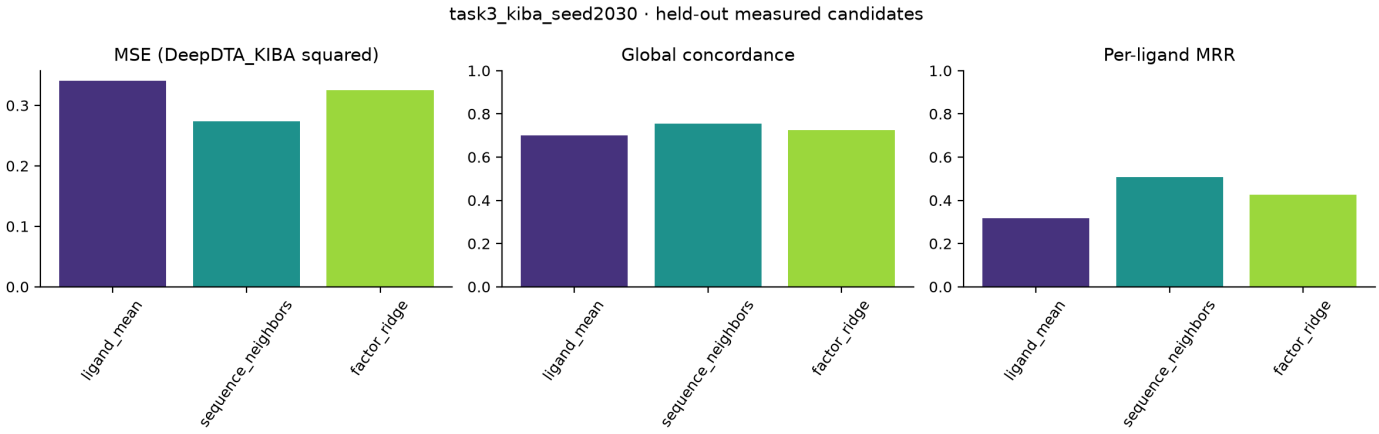


KIBA, seed 2029 — cold-target model comparison

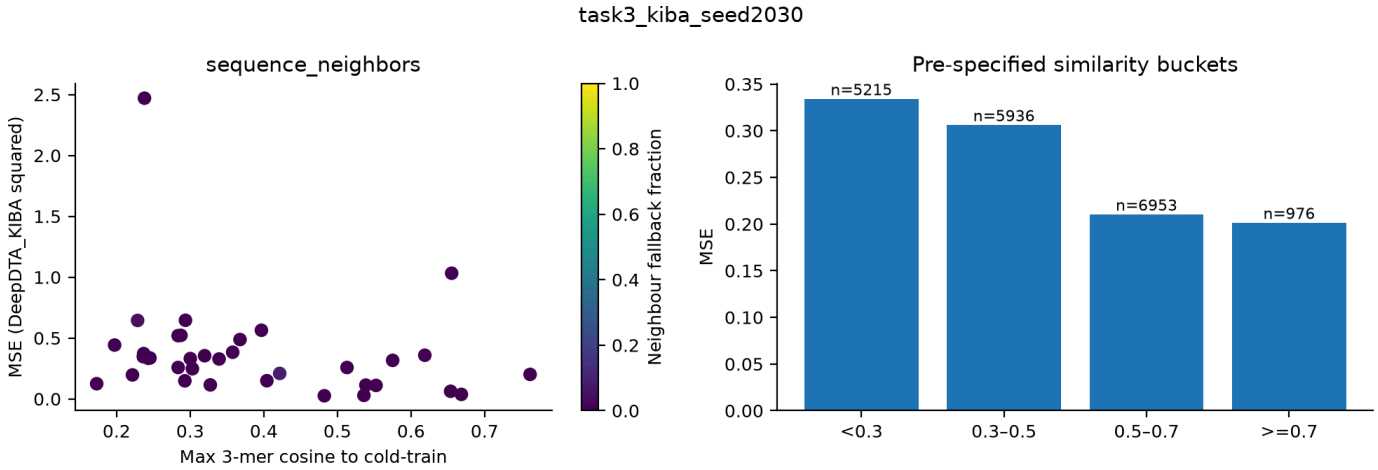


KIBA, seed 2029 — similarity and error diagnostics

Task 3 figures — KIBA, seed 2030



KIBA, seed 2030 — cold-target model comparison



KIBA, seed 2030 — similarity and error diagnostics